Artificial Intelligence

A Brief History of AI

- During the *Second World War*, noted British computer scientist ***Alan Turing*** worked to crack the 'Enigma' code which was used by German forces to send messages securely.
- Alan Turing and his team created the Bombe machine that was used to decipher Enigma's messages.

A Brief History of AI

- The Enigma and Bombe Machines laid the foundations for Machine Learning.

A Brief History of AI

- According to Turing, a machine that could converse with humans without the humans knowing that it is a machine would win the “imitation game” and could be said to be “intelligent”

A Brief History of AI

- In 1956, American computer scientist John McCarthy organised the Dartmouth Conference, at which the term '***Artificial Intelligence***' was first adopted

Getting Serious About AI Research

- In the 1960s, researchers emphasized developing algorithms to solve mathematical problems and geometrical theorems.

Getting Serious About AI Research

- In the late 1960s, computer scientists worked on Machine Vision Learning
- Developed machine learning in robots.
- WABOT-1, the first ‘intelligent’ humanoid robot, was built in Japan in 1972.

Getting Serious About AI Research

AI Winters

- To be successful, AI applications (such as vision learning) required the processing of enormous amount of data.
- Computers were not well-developed enough to process such a large magnitude of data.
- ***Governments and corporations were losing faith in AI.***

AI Winters

- Therefore, from the mid 1970s to the mid 1990s, computer scientists dealt with an acute shortage of funding for AI research.

New Era of AI

- In the late 1990s, American corporations once again became interested in AI.
- In 1997, IBM's Deep Blue defeated and ***became the first computer to beat a reigning world chess champion,*** Garry Kasparov

Deep Blue v/s Garry Kasparov

New Era of AI

- Exponential gains in computer processing power and storage ability.
- In the past 15 years, Amazon, Facebook, Google, Baidu, and others leveraged machine learning to their huge commercial advantage.

Artificial Intelligence

A branch of Computer Science. Examines how we can **achieve intelligent behavior** through computation.

Artificial Intelligence

What actually the intelligent behavior is?

What is intelligence?

Are these intelligent?

What is intelligence?

What about these?

What is intelligence?

Webster says:

- *The capacity to acquire and apply knowledge.*
- *The faculty of thought and reason.*

What is intelligence?

What **features/abilities** do humans (animals/animate objects) have that you think are indicative or characteristic of **intelligence**?

Abstract concepts, mathematics, language, problem solving, memory, logical reasoning, planning ahead, emotions, morality, ability to learn/adapt, etc…

What is intelligence?

What **features/abilities** do humans (animals/animate objects) have that you think are indicative or characteristic of **intelligence**?

Today’s Lecture

- Weak and Strong AI
- Acting humanly
- Think like humans
- think rationally
- Acting rationally
- Turing Test
- Chinese Room Argument

What is AI?

The exciting new effort to make computers thinks … *machine with minds*, in the full and literal sense" (Haugeland 1985)
The automation of activities that we associate with human thinking, activities such as decision-making, problem solving, learning ..." (Bellman, 1978)

"The study of mental faculties through the use of computational models" (Charniak et al. 1985)
The study of the computations that make it possible to perceive, reason, and act" (Winston, 1992)

"The art of creating machines that perform functions that require intelligence when performed by people" (Kurzweil, 1990)
The study of how to make computers do things at which, at the moment, people are better" (Rich and Knight, 1991)

A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes" (Schalkol, 1990)
The branch of computer science that is concerned with the automation of intelligent behavior" (Luger and Stubblefield, 1993)

What is AI?

ARTIFICIAL INTELLIGENCE

Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

Weak and Strong AI

- Weak AI
  - Computers can be programmed to act **as if** they were intelligent (as if they **were** thinking)
- Strong AI
  - Computers can be programmed to think (i.e. they **really** are thinking)

Weak and Strong AI

- Weak AI is AI that can not 'think', i.e. a computer chess playing AI does not think about its next move, it is based on the programming it was given, and its moves depend on the moves of the human opponent.

- Strong AI is the idea/concept that we will one day create AI that *can* 'think' i.e. be able to play a chess game that is not based on the moves of the human opponent or programming, but based on the AI's own 'thoughts' and feelings and such, which are all supposed to be exactly like a real humans thoughts and emotions and stuff.

What is AI?

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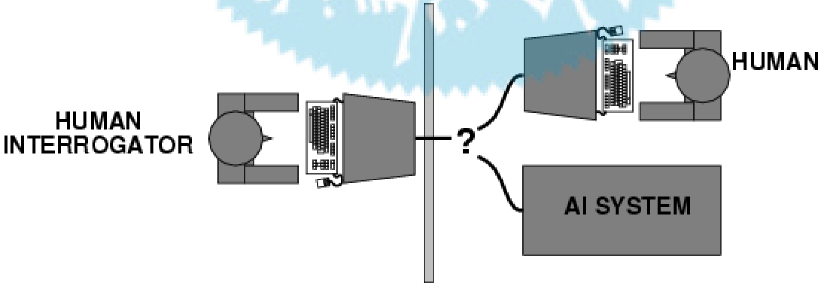
Acting humanly

The Turing Test approach

- Turing (1950) "Computing machinery and intelligence"
- The Turing Test



Alan Turing

Alan Turing, born at 23rd of june, 1912

“I believe that in about fifty years’ time it will be possible to programme computers, with a storage capacity of about 10^9, to make them play the imitation game so well that an average interrogator will not have more than 70 percent chance of making the right identification after5 minutes of questioning”

-Alan Turing (1950)

Turing Test

- *“Turing was convinced that if a computer could do all mathematical operations, it could also do anything a person can do“*
- **Computing Machinery and Intelligence**, written by Alan Turing and published in 1950 in *Mind*, is a paper on the topic of artificial intelligence in which the concept of what is now known as the Turing test was introduced to a wide audience.

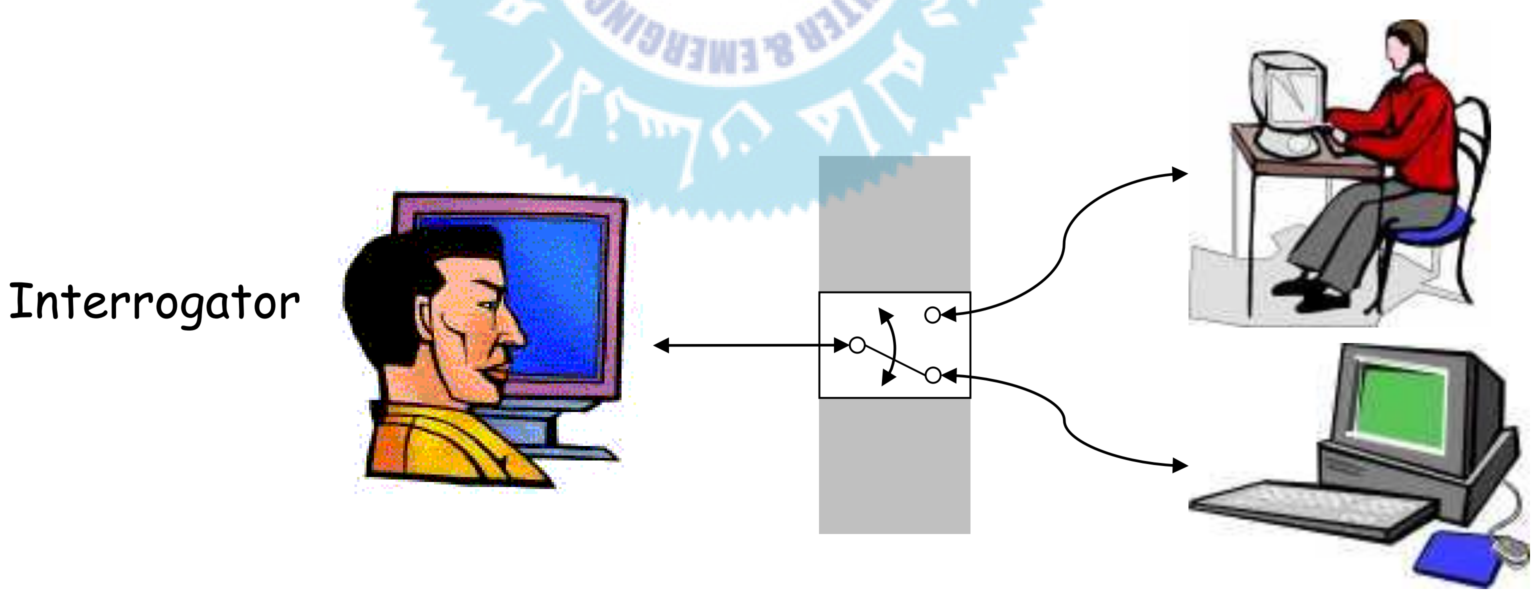
The Turing Test

- Today the Game is usually referred to as the Turing Test.
- If a computer can play the game just as well as a human, then the computer is said to ‘pass’ the ‘test’, and shall be declared intelligent.

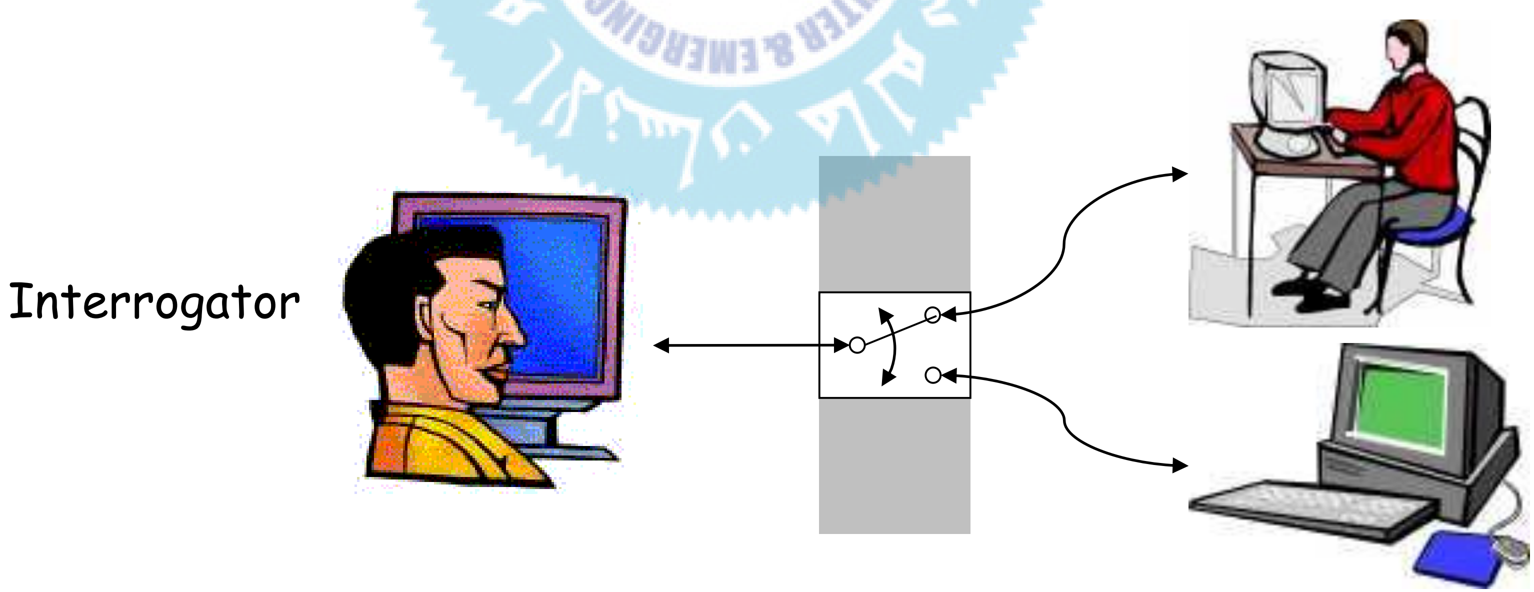
Turing Test

- **How can we evaluate intelligence?**
  - Turing [1950]: a machine can be deemed intelligent when its responses to interrogation by a human are indistinguishable from those of a human being.

Turing Test



Turing Test



Total Turing Test

- includes a video signal so that the interrogator can test the subject's perceptual abilities, as well as the opportunity for the interrogator to pass physical objects ``through the hatch.''
- To pass the total Turing Test, the computer will need
  - **computer vision** to perceive objects, and
  - **robotics** to move them about.

Turing Test

“On the internet, nobody knows you're a dog.”

How effective is this test?

- Agent must:
  - Have command of language
  - Have wide range of knowledge
  - Demonstrate human behavior (humor, emotion)
  - Be able to reason
  - Be able to learn
- Loebner prize competition is modern version of Turing Test
  - (The **Loebner Prize** is an annual competition in artificial intelligence that awards prizes to the chatterbot considered by the judges to be the most human-like.)
  - Example: Alice, Loebner prize winner for 2000 and 2001

Turing Test: Criticism

- What are some potential problems with the Turing Test?
  - Some human behavior is not intelligent
    - the temptation to lie, a high frequency of typing mistakes
  - Some intelligent behavior may not be human
    - If it were to solve a computational problem that is practically impossible for a human to solve
  - Human observers may be easy to fool
    - A lot depends on expectations
    - Chatbots, e.g., ELIZA, ALICE
  - Chinese room argument
- Is passing the Turing test a good scientific/engineering goal?

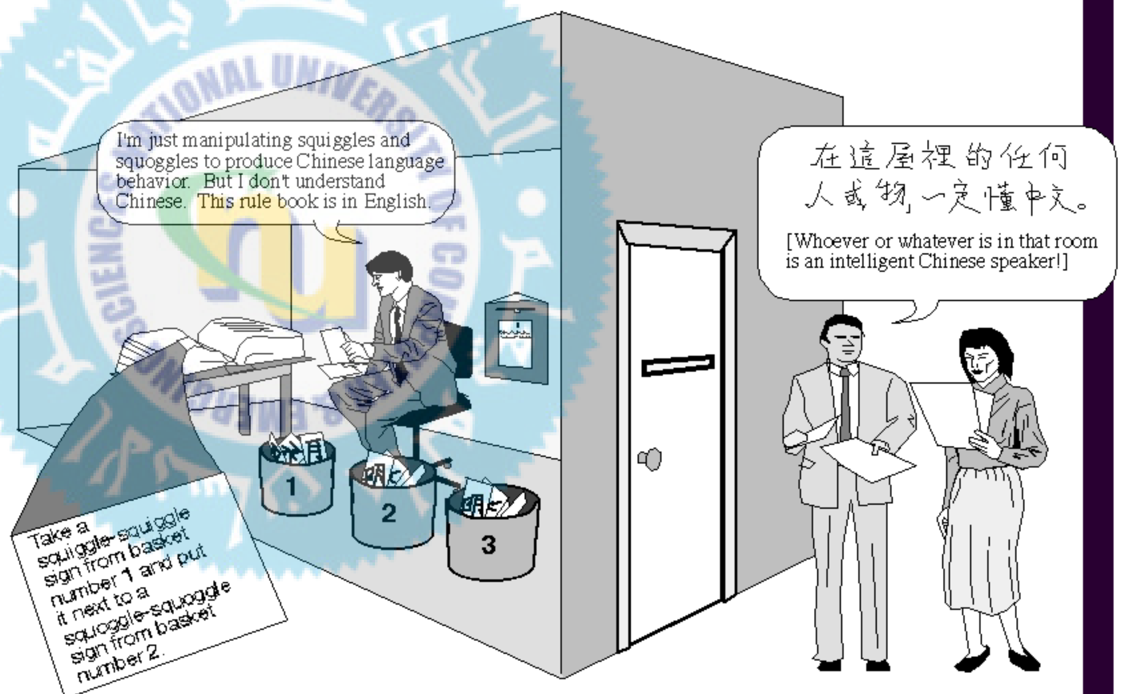
Chinese Room Argument

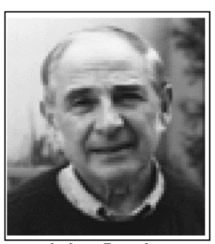
- Devised by John Searle
- An argument against the possibility of true artificial intelligence.

Chinese Room Argument

3 John Searle, 1980a, 1980b, 1990b

The Chinese Room argument. Imagine that a man who does not speak Chinese sits in a room and is passed Chinese symbols through a slot in the door. To him, the symbols are just so many squiggles and squoggles. But he reads an English-language rule book that tells him how to manipulate the symbols and which ones to send back out. To the Chinese speakers outside, whoever (or whatever) is in the room is carrying on an intelligent conversation. But the man in the Chinese Room does not understand Chinese; he is merely manipulating symbols according to a rule book. He is instantiating a formal program, which passes the Turing test for intelligence, but nevertheless he does not understand Chinese. This shows that instantiation of a formal program is not enough to produce semantic understanding or intentionality. **Note:** For more on Turing tests, see Map 2. For more on formal programs and instantiation, see the "Is the brain a computer?" arguments on Map 1, the "Can functional states generate consciousness?" arguments on Map 6, and sidebar, "Formal Systems: An Overview," on Map 7.





John Searle

in · ten · tion · al · it · y: The property (in reference to a mental state) of being directed at a state of affairs in the world. For example, the belief that Sally is in front of me is directed at a person, Sally, in the world. Intentionality is sometimes taken to be synonymous with representation, understanding, consciousness, meaning, and semantics. Although there are important and subtle distinctions in the definitions of "intentionality," "understanding," "semantics," and "meaning," in this debate they are sometimes used synonymously.

Chinese Room Argument

"The reason that no computer program can ever be a mind is simply that a computer program is only syntactical, and minds are more than syntactical. Minds are semantical, they have content." - John Searle

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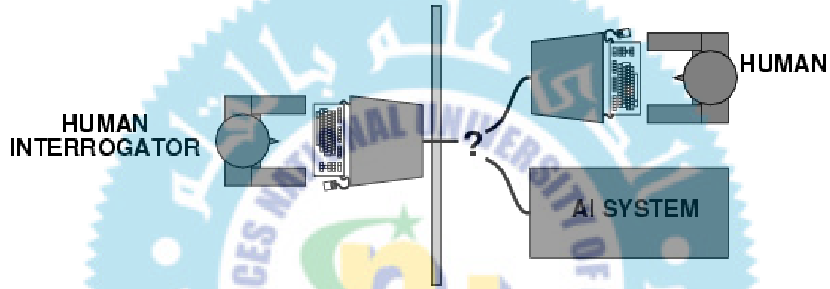
- 

- What capabilities would a computer need to have to pass the Turing Test?
  - Natural language processing
  - Knowledge representation
  - Automated reasoning
  - Machine learning
- Turing predicted that by the year 2000, machines would be able to fool 30% of human judges for five minutes

Thinking humanly
The cognitive modeling approach

- Goal: Develop precise theories of human thinking
- Cognitive Architecture
  - Software Architecture for modeling human performance
  - Describe task, required knowledge, major sub-goals
  - Architecture follows human-like reasoning
  - Makes testable predictions: Time delays during problem solving, kinds of mistakes, eye movements, verbal protocols, learning rates, strategy shifts over time, etc.
- Problems:
  - It may be impossible to identify the detailed structure of human problem solving using only externally-available data.

Thinking humanly

The cognitive modelling approach

- We need to **get *inside* the actual workings** of human minds.
- There are two ways to do this: through
  - trying **to catch our own thoughts as** they go by
  - or through psychological experiments.
- Cognitive science: the brain as an *information processing machine*
  - Requires scientific theories of *how the brain works*
- How to understand cognition as a computational process?
  - try to think about how we think
  - Predict and test behavior of human subjects
  - Image the brain, record neurons
- The latter two methodologies are the domains of cognitive science and cognitive neuroscience

Thinking rationally

The laws of thought approach

- Idealized or "right" way of thinking
- **Logic:** *patterns of argument that always yield correct conclusions when supplied with correct premises*
  - "Tom is a man; all men are mortal; therefore Tom is mortal."
- Beginning with Aristotle, philosophers and mathematicians have attempted to formalize the rules of logical thought
- ***Logicist* approach to AI:** describe problem in formal logical notation and apply *general deduction procedures* to solve it
- Problems with the logicist approach
  - Computational complexity of finding the solution
  - Describing real-world problems and knowledge in logical notation
  - Dealing with uncertainty
  - A lot of intelligent or "rational" behavior has nothing to do with logic

Thinking Rationally: The Logical Approach

- Ensure that all actions performed by computer are justifiable ("rational")

Facts and Rules in Formal Logic ⟷ Theorem Prover

- *Rational = Conclusions are provable from inputs and prior knowledge*
- Problems:
  - Representation of informal knowledge is difficulty
  - Hard to define "provable" reasoning

Acting rationally
Rational agent

- A rational agent is one that acts to achieve the best expected outcome
  - Goals are application-dependent and are expressed in terms of the **utility of outcomes**
  - Being rational means ***maximizing your expected utility***
  - In practice, utility optimization is subject to the agent's computational constraints

Acting Rationally
Rational Agents

- Claim: "Rational" means more than just logically justified. *It also means "doing the right thing"*

> *Rational agents do the best they can given their resources*

Weak and Strong AI

Strong AI	Weak AI
Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

Summery of Today's Lecture

- Weak and Strong AI
- Acting humanly
- Think like humans
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- Turing Test
- Chinese Room Argument